

Graphics chipsets have undergone a major metamorphosis from the time they were first introduced. Manufacturers have always kept the end-users interested by releasing newer, faster and even more feature-crammed chipsets. The graphics chipset has evolved from a polygon-pumping machinery to a completely programmable solution that could very well revolutionise the way we work and play.

It all started innocently enough when 3Dfx released their Voodoo chipset and forever changed the way games were perceived. Things soon snowballed and the year 1998 saw a boom in the graphics sector with the release of a number of high performance 3D chipsets like ATI's Rage 128, nVidia's Riva TNT, Matrox's G200 and S3's Savage 3D.

The intense competition forced manufacturers to adopt a six-monthly product cycle that doubled the performance of their offerings. The heat of the game got to a few companies and they had to leave the business, huffing and panting—3Dfx made a surprising exit, Matrox went the 2D way, 3DLabs decided to cater to the niche market of CAD/CAM professionals. The gamer got the short end of the stick with only ATI and nVidia left standing in the fray.

What followed was a lot of catch-me-if-you-can between nVidia and ATI, as both tried to go one-up on the other with cyclical assaults. nVidia's GeForce and GeForce2 were countered by ATI's Radeon chipset, the GeForce3 GPU went up against the Radeon 8500, which was in turn upstaged by the GeForce4.

As things currently stand, nVidia still rules the market thanks to their aggressive release cycles and their excellent driver quality. But before you take the leap of faith and purchase the latest video card, temper that decision with what the Indian market has to offer you and get a better picture on the chipset that powers your next graphics card.

We have tested a number of graphics cards based on 15 discrete graphics chipsets currently available in India, to help you make that all-important buying decision.

## Market reality in India

A 3D graphics card is still considered a luxury by most computer users in India and the market is still nascent. Product

launch times are skewed to such an extent in India that by the time you get your hands on a GeForce4 card, nVidia would already be ready to launch its successor. Timing apart, the price at which the cards are initially sold is simply unjustifiable. This is the primary reason that compels users to shy away from high-end cards. When a graphics card is imported into India, many surcharges and custom duties are applied on it. Additionally, the local dealer and distributor knows the relative low volumes the product will sell and the short life span of each high-end graphics chipset, which forces him to levy more margins on the product, taking it further out of the reach of most users.

Also even though almost all the high-end chipsets are available in the Indian market, a major chunk of the users swear by their outdated SiS and nVidia TNT2 M64 based cards. The fact is that there are a number of better performing chipsets available at the same price point, but users are simply not aware of the situation. Take for example, the ATi Radeon VE and the nVidia GeForce2 MX200/MX400—the cards sporting these chipsets provide decent performance, are a great value for money and are ideally suited for the budget audience.

## Digit Test Process

We tested cards based on all the available graphics chipsets in the market. Since most card manufacturers closely follow the chipset vendor's listed specifications, the real-world performance difference between a GeForce card from Vendor X and Vendor Y is negligible. What's important is having the requisite information to choose the graphics chipset that is right for you.

The test system consisted of an Intel Pentium 4 2.4 GHz processor based on the Intel D850EMV2 motherboard along with 256 MB of RDRAM and a 20 GB, 7,200-rpm hard disk. All 3D tests were conducted under Windows 98SE and we used Windows XP for testing the 2D performance of the cards. Prior to running the tests on every card, a ghost image of Windows 98SE was loaded, followed by a driver update. The desktop resolution was kept constant at 1024x768x16 at a refresh rate of 75 Hz. DirectX version 8.1 was used. VSync was turned off through the Display Properties and the 'Always on top' mode was disabled for the Windows taskbar.

## Something for Everyone

Graphics cards don't come in the plain vanilla flavour anymore; there are many different versions available for each graphics chipset, each at varying price points. Thus, a GeForce2 chipset will have an MX variant for the budget crowd and an Ultra version for those that demand performance. Also, manufacturers add funny sounding names like Ultra, Pure, Deluxe, Pro, VIVO, etc to their cards to highlight additional features such as higher memory speeds, TV tuner capability, FM radio, Video Out, VR glasses, etc. As an example, take the cards based on the GeForce2 MX200 chipset. The Maxtone GeForce2 MX200 based cards come in three flavours: one features a TV-Out port (Rs 1,900), while the other doesn't (Rs 1,500); both have 32 MB of on-board RAM. The third variant comes with both TV-Out and an S-Video port and is priced at Rs 2,400. Sounds confusing? It is!

For testing the onboard video chipsets we used the following test beds:

- Intel 815 using the Intel D815EEA2 motherboard with a Pentium III Tualatin 1.13 GHz and 256 MB SDRAM
- Intel 845GL using the Intel D845GBV motherboard with a Pentium 4 2.4GHz and 256 MB DDR RAM

## Test method

We evaluated the different chipsets on performance (75 per cent weightage) and value for money (25 per cent weightage). To separate their offering from the crowd, vendors routinely offer features such as video-in/video-out or support for 3D virtual glasses. Our standard parameters such as features, and warranty and support do not apply here since we're evaluating the graphics chips themselves and not cards based on these solutions.

## Performance

When one thinks of a graphics chipset, the first question that comes to our mind is: How powerful is it? The power of a 3D graphics chipset is mainly reflected by the average frames per second it is able to output and the new technologies that it sports to give rich visual quality at sustained high frame rates. Therefore, our benchmarks consist of a number of speed and quality tests. The overall weightage